INCA – Intergalactic Number Conversion Academy

This document contains the **final, optimized slide-by-slide content** for your university project presentation, including: - Clear, simplified slide text (audience-friendly) - What to edit or remove - Exact screenshot filenames to include - Suggestions on adding or replacing slides

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Slide 1 – Title & Team

Title: Intergalactic Number Conversion Academy (INCA)

Subtitle: A sci-fi themed learning tool for number systems (Python Desktop Application)

Module: Mathematics for Computing

Team Members: (list names)

Roles (one line): UI & Story | Algorithms | Practice Mode | Testing & Documentation

Screenshot: - IMG_0193.JPG (Welcome popup – cropped)

Slide 2 – Problem & Motivation

The Problem We Solved - Number system concepts are often confusing and boring for students - Most tools only show the final answer, not the working steps - Lack of engagement reduces understanding and motivation

Our Goal - Make number systems easy, visual, and fun using game-based learning

Screenshot (optional): - 20.JPG (Mission Simulator – small image)

Slide 3 – Project Overview

What is INCA? - A Python desktop application that teaches number systems interactively - Uses a sci-fi academy theme to guide learners - Includes practice missions, step-by-step explanations, and a final exam

Why it works - Converts theory into hands-on problem solving

Screenshot: - IMG_0194.JPG (Main dashboard / navigation panel)

Slide 4 – Mathematical Foundations

Concepts Covered - Number systems: Binary, Octal, Decimal, Hexadecimal - Conversion techniques: - Repeated division - Place-value expansion - Signed binary representations: - 1’s complement - 2’s complement - Binary arithmetic (addition & subtraction)

Screenshots: - 21.JPG (2’s complement arithmetic with negative number – main) - 28.JPG (Base arithmetic with binary result – optional)

Slide 5 – Computing Concepts & Tools

How INCA Was Built - Python for logic and calculations - Tkinter for desktop graphical interface - Modular design: - Signal Decoder - Arithmetic Engine - Complement Converters - Mission Simulator

Quality & Safety - Input validation and error handling to prevent incorrect usage

Screenshot (optional): - 23.JPG (Validation / error popup – small)

Slide 6 – Key Features

Key Features of INCA - Step-by-step explanations instead of only final answers - Interactive practice missions with scoring - Built-in training manual for reference - Sci-fi themed interface for engagement

Screenshots: - 20.JPG (Mission Simulator – main image) - 33.JPG (Training Manual – secondary image)

Slide 7 – Story & Gamification

Why INCA is Fun - User plays as a cadet in the Intergalactic Academy - Alien signals must be decoded using number systems - Completing missions increases credibility and progress - Learning happens naturally through gameplay

Screenshot: - IMG_0193.JPG (Welcome / story popup)

Slide 8 – Demo Flow (Recommended Replacement)

Live Demonstration Flow 1. Start at INCA dashboard 2. Decode a signal using base conversion 3. Use arithmetic engine (add/subtract across bases) 4. Perform 2’s complement arithmetic 5. Enter mission simulator and view progress 6. Show training manual and final exam

Screenshot: - IMG_0194.JPG or 20.JPG

Slide 9 – Conclusion & Future Work

Conclusion - INCA makes number systems easier through interactive learning - Step-by-step logic improves understanding - Gamification keeps students motivated

Future Improvements - More difficulty levels and bases - Leaderboard and saved progress - Expanded quiz system

Screenshot (optional): - 20.JPG (small)

Recommended Addition

New Slide – Team Contribution

- Member A: UI & theme design
- Member B: Base conversion algorithms
- Member C: Arithmetic engine
- Member D: Mission simulator & scoring
- Member E: Testing & documentation

Final Notes

- Total slides with images: 9–10 (ideal)
- Avoid duplicate screenshots
- Let visuals explain; keep text minimal
- Focus presentation on *how it works* and *why it helps learning*